SYNERGETIC OPERATING SYSTEM (SOS)

A Topology-Based, Emergent, and Resource-Field Operating System Architecture

Inventor / Originator

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GitHub: https://github.com/solomonyaw
Project: https://github.com/solomonyaw/Synergetics-Operating-System

📄 **Whitepaper**:
https://github.com/solomonyaw/Synergetics-Operating-System/blob/master/SYNERGETIC%20OPERATING%20SYSTEM%20-%20White%20Paper.pdf

📄 **Simulation Code**: https://github.com/solomonyaw/Synergetics-Operating-System/blob/master/synergetic_scheduler_rust_simulation.rs

ABSTRACT

This document introduces the **Synergetic Operating System (SOS)**, a novel operating system architecture based on **topological computation, relational process modeling, and emergent resource optimization**. Unlike conventional operating systems that rely on hierarchical scheduling and deterministic resource allocation, SOS models computation as a **dynamic graph embedded in a resource field**, where execution arises from systemic relationships rather than isolated process control.

The system is inspired by principles of Synergetics as developed by Buckminster Fuller, emphasizing whole-system efficiency, relational geometry, and ephemeral resource utilization.

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1. TECHNICAL FIELD

This invention relates to:

- Operating system design
- Distributed computing architectures
- Scheduling algorithms
- Resource management systems
- Graph-based computation models
- Adaptive and self-organizing systems

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2. BACKGROUND

Traditional operating systems (e.g., Linux, Windows, macOS) rely on:

- Process hierarchies (tree structures)
- Time-sliced CPU scheduling
- Static resource abstraction layers
- Deterministic execution models

These systems face limitations:

- Inefficient global optimization
- High overhead in resource isolation
- Limited adaptability to dynamic workloads

- Lack of structural emergence in computation

There is a need for a system that:

Treats computation as a **relational, adaptive, and emergent field phenomenon rather than isolated processes**

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3. SUMMARY OF THE INVENTION

The Synergetic Operating System introduces:

Core Innovation:

A **Topology-Based Execution Model (TBEM)** in which:

- Processes are represented as **nodes in a dynamic graph**
- Interactions are represented as **weighted edges**
- System state exists as a **resource-energy field**
- Execution is determined by **global optimization of relational structure**

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4. SYSTEM ARCHITECTURE

4.1 Topological Process Model

Each process is defined as:

[
$P_i = (R_i, C_i, L_i, E_i)$
]

Where:

- (R_i) = resource demand vector
- (C_i) = connectivity degree
- (L_i) = locality score
- (E_i) = emergent impact factor

Processes are nodes in a graph:

[
G = (V, E)
]

4.2 Resource Field Abstraction

Instead of discrete CPU/memory separation:

[
F_{resource} = {CPU, Memory, IO, Cache}
]

is treated as a **continuous field**, dynamically allocated based on system pressure gradients.

4.3 Synergetic Scheduler

Scheduling is defined as:

[
S(P_i) = \alpha R_i + \beta C_i + \gamma L_i + \delta E_i - \epsilon D_i
]

Execution selects:

[
P_{next} = \arg\max S(P_i)
]

This replaces time-slicing with **relational optimization scheduling**.

4.4 Emergent Optimization Layer

System continuously evaluates:

- Graph topology evolution
- Resource redistribution efficiency
- Execution feedback loops

Result:

System behavior emerges from interactions, not predefined schedules

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4.5 Ephemeral Execution Model

Processes are:

- Non-persistent unless reinforced by system demand
- Automatically decayed if not contributing to system efficiency

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5. DETAILED DESCRIPTION OF OPERATION

Step 1: Process Graph Construction

- Processes are instantiated as nodes
- Inter-process communication forms weighted edges

Step 2: System State Evaluation

- Compute global graph metrics
- Evaluate resource field pressure

Step 3: Scheduling Decision

- Compute score function for each node
- Select highest scoring node

Step 4: Execution Dispatch

- Allocate system resources dynamically
- Execute process for bounded interval

Step 5: Feedback Update

- Update node weights
- Reinforce or decay edges
- Adjust resource field distribution

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6. CLAIMS (PATENT-STYLE)

Claim 1

A computing system comprising:

- a graph-based representation of processes
- a resource field abstraction model
- a topology-aware scheduling algorithm
  where execution is determined by relational scoring rather than fixed priority queues.

Claim 2

The system of Claim 1, wherein process scheduling is computed using a multi-variable emergent score function incorporating:

- resource demand
- graph connectivity
- locality metrics
- emergent system contribution

Claim 3

A system wherein computational resources are represented as a continuous field rather than discrete partitions.

Claim 4

A method of operating a computing system wherein processes dynamically form and dissolve based on system-wide optimization feedback.

Claim 5

A system wherein scheduling decisions are based on global graph optimization rather than local process priority.

Claim 6

A system capable of self-restructuring its process topology based on runtime interaction patterns.

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7. NOVELTY STATEMENT

The Synergetic Operating System introduces the following novel elements:

1. **Graph-native operating system architecture**
2. **Resource field computation model**
3. **Emergent scheduling algorithm**
4. **Ephemeral process lifecycle management**
5. **Topology-driven execution control**
6. **System-wide relational optimization engine**

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8. ADVANTAGES OVER PRIOR ART

Feature	Traditional OS	Synergetic OS
Process model	Hierarchical	Graph-based
Scheduling	Time-slice	Emergent optimization
Resource model	Static allocation	Continuous field
Optimization scope	Local	Global
Process lifecycle	Persistent	Ephemeral

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9. IMPLEMENTATION MODES

Mode A: Simulation Layer

- Rust or Python graph simulation

Mode B: Hybrid Kernel Extension

- Integration into Linux kernel scheduler layer

Mode C: Full Kernel Implementation

- Rust-based custom microkernel

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10. INDUSTRIAL APPLICATIONS

- Cloud computing optimization
- AI-native operating systems
- Edge computing orchestration
- Distributed compute fabrics
- Autonomous system infrastructure

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11. INVENTORSHIP AND PRIORITY CLAIM

The Synergetic Operating System is the original conceptual invention of:

Solomon Yaw Adeklo

This document establishes:

- priority of conceptual authorship
- originality of system architecture
- record of technical disclosure

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12. CONCLUSION

The Synergetic Operating System represents a paradigm shift from:

- deterministic computation → emergent computation
- hierarchical control → topological control
- static resource allocation → dynamic resource fields

This architecture enables a new class of operating systems where:

system intelligence emerges from relational structure rather than predefined control logic.